

# ACSD: Authorized Lineage for Anonymous Scholarly Releases

Anonymous authors

## Abstract

Anonymous scholarly disclosure needs a more precise answer than “put a signed PDF in a repository.” A signature identifies a signing key, not necessarily a person; a repository timestamp is not independent time; and an exact predecessor hash does not say who may publish the next version. This paper presents ACSD, a narrow executable profile for anonymous scholarly releases and authorized lineage. A compact approval target binds exact release, governance, provenance-policy, and lineage-transition digests before every required pseudonymous key signs it; a second acyclic object closes over the exact signature bytes before optional timestamping. An unchanged predecessor authority can authorize an exact child through those same approvals, while an authority change additionally requires predecessor-quorum signatures over the exact transition. A typed evidence-to-claim rule supports atomic event opening, partial or full per-slot unblinding, and publication crosswalks without silently granting a different claim. The optional series layer remains separate, permitting cyclic semantic citation while exact commitments stay acyclic. Evaluation includes fresh-key  $n+1$  and timestamp-completion attacks, a non-strawman same-primitive lineage comparison, one-to-three-slot unblinding, cross-work key-linkability checks, 64 fixed and 1,000 generated Python–Node canonical-JSON cases, 163 Python tests, and an 84-theorem Lean development whose restricted appraisal-transcript decoder agrees with Python on 19 production cases. None of these mechanisms proves natural-person identity, contribution truth, originality, peer review, venue acceptance, or legal nonrepudiation.

## 1 Introduction

Researchers often want to disclose a manuscript before conventional publication while keeping the author identity unavailable during review. A common informal proposal is to upload an anonymous file, sign it, and rely on repository history. That proposal leaves several distinct questions entangled.

One concrete trigger is the interval around an arXiv submission: a first-time or new-category submission may require endorsement, while moderation and review are separate from scientific peer review. During that interval an author may want a public, content-anonymous evidence package that preserves exact bytes and team assent without pretending to replace arXiv’s process. We use this as an application scenario, not as a claim that arXiv queues have a measured backlog or that ACSD should circumvent its rules.

1. What exact bytes did one or more keys approve?
2. Can a later verifier distinguish a revision, a legitimate branch, and an equivocation?
3. Can a sequence of papers cite one another, including mutual citations, without a circular content-hash dependency?
4. What does a time witness establish, and what does it not establish?
5. Can later event or identity disclosure remain limited to its exact authorized scope?

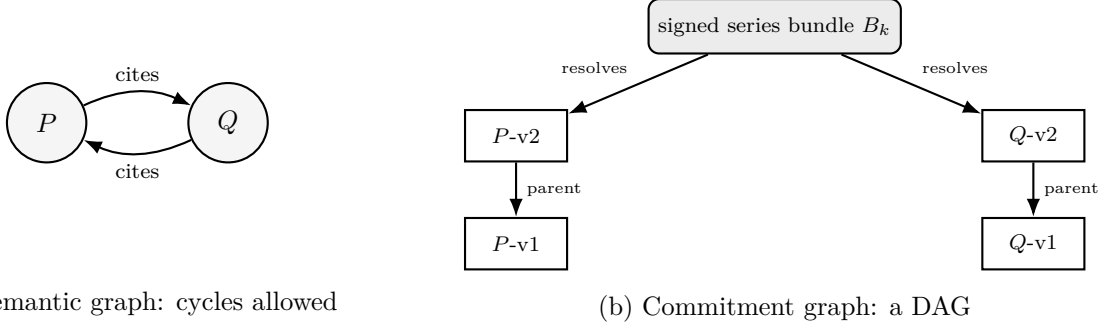


Figure 1: ACS D separates cyclic semantic citations over stable WorkIDs (a) from acyclic exact-byte commitments through release digests and predecessor links (b). Commitment arrows point toward the committed object and indicate neither chronology nor priority.

The first question is about key control and integrity. It is not, by itself, a proof that a particular natural person wrote the work. The second and third are data-model and verification questions. The fourth requires an external witness such as an RFC 3161 timestamp authority (TSA), not a local clock or a mutable Git field. Conflating these layers produces systems that overclaim.

This paper presents ACS D (Anonymous Scholarly Claim and Disclosure), a deliberately narrow application profile. It uses standard content hashing, COSE signatures, and provenance concepts rather than proposing a new primitive. The profile is designed for content-anonymous release: the public paper and artifact do not name authors, but an account hosting them can still be linkable. It is therefore *not* described as author-unlinkable double blind publication.

The main contributions are:

1. a self-contained paper-release object with ordered anonymous author slots and per-author endorsements;
2. a two-layer construction that permits cyclic scholarly citations over stable WorkIDs while keeping content commitments acyclic;
3. signed UTF-8 byte-range citation witnesses and a restricted canonical JSON profile to reduce field-interpretation ambiguity;
4. explicit classifications for branches, equivocation, and copied content under a fresh lineage;
5. an acyclic unanimous-approval target and closed capability calculus that prevent unsigned policy amplification and approval reuse after governance or PEC substitution;
6. an exact lineage transition requiring predecessor-quorum signatures, rejecting a fresh-key  $n+1$  claimant while supporting explicit author/key changes and branches;
7. a signature-complete approval-set timestamp subject and a small typed evidence-composition core for atomic event disclosure, partial/full per-slot unblinding, and exact publication cross-walks, including an evidence-union non-amplification theorem; and
8. an executable prototype with one-command release, revision, and unblinding paths, a cross-work key-linkability audit, independent COSE verification, adversarial RFC 3161 checks, generated differential testing, a controlled adjacent-signature comparison, and a production appraisal transcript checked by both the executable and the Lean composition model.

## 2 Scope and Threat Model

**Assets.** A release contains manuscript bytes, an ordered list of anonymous author keys, roles and contribution declarations, an AI-use declaration, and optional parent-release information. A series package contains references to completed releases, a series key, sequencing information, and cross-paper semantic relations. Public keys are pseudonymous identifiers.

**Adversary.** We consider a party that can copy public content, generate fresh keys, reuse the exact WorkID and predecessor digests, claim the next unused version number, reorder or substitute unsigned metadata, timestamp a target before adding signatures, replay a valid disclosure under another scope, present a partial byline as complete, reuse one public key across unrelated works, present a signed but malformed JSON payload, and claim unsupported cross-paper citations. We also consider a series key that signs two incompatible objects. We do not assume that a signature reveals a human identity, that a contribution declaration is factually true, or that a local timestamp is trusted.

**Non-goals.** ACSD does not prove authorship by a natural person, independent discovery, plagiarism, venue acceptance, peer review, or legal nonrepudiation. These are not merely unimplemented features: some require social, institutional, or legal evidence outside the cryptographic object. The profile instead preserves evidence that can be useful in a later dispute.

## 3 Release Layer

### 3.1 Canonical release object

Each *PaperRelease* has a random stable WorkID, a visible slot (*WorkID*, *version*, *line*), a content path, content length and digest, ordered author records, declared roles, contributions, an AI-use declaration, optional exact predecessor, and a list of citation witnesses. The release identifier is

$$\text{ReleaseID} = \text{urn:sha256:H}(\text{canonical}(R)).$$

The content digest is over exact manuscript bytes. The release object uses a deliberately restricted JSON encoding: recursively sorted ASCII object keys, no insignificant whitespace, and safe integers only. A verifier parses, re-canonicalizes, and compares bytes before accepting a signature. This is a profile rule, not a claim that it is a general JSON standard.

### 3.2 Joint approval target and closed capability calculus

The *AuthorshipGovernanceStatement* is the exact declaration of the work id, manuscript digest, ordered byline slots and roles, corresponding-author key, and AI-use field. The CLI derives it from the team configuration and release; there is no privileged coordinator whose unsigned word makes it authoritative. Instead, every declared author key approves its digest through the common target described below. This proves assent to the declaration, not the factual truth of its roles or contribution history.

A *Provenance Evidence Capsule* (PEC) binds the release subject and governance digest to a digest-linked evidence-event sequence and a claim policy. The policy contains a fixed-vocabulary list of permitted outcomes, their required capabilities, and mandatory non-claims. A PEC also names one team key as an issuer for provenance, but that label alone grants no outcome: the complete team must still approve the common target.

```
{  
  "schema": "acsd-approval-target/v2",  
  "work_id": "urn:uuid:12345678-1234-1234-1234-123456789abc",  
  "release_digest": "111111111111111111111111111111111111111111111111111111111111111111",  
  "governance_digest": "222222222222222222222222222222222222222222222222222222222222222222",  
  "pec_digest": "333333333333333333333333333333333333333333333333333333333333333333",  
  "required_key_ids": ["444444444444444444444444444444444444444444444444444444444444444444"],  
  "lineage_transition_digest": null  
}
```

$$\text{Granted}(P, o) \iff \text{Accepted}(P) \wedge o \in P.\text{permittedOutcomes} \wedge \text{CapabilityEstablished}(P, o).$$

Without the joint target, suppose authors sign only the release while an attacker later changes the unsigned governance object or PEC policy. The attacker could relabel roles, add the external-time outcome, or try to map a local clock to that outcome while the old release signatures still verify. Under ACS<sub>D</sub>, changing the governance or policy changes its digest and causes a target-binding failure; recomputing the target instead causes the old COSE approvals to fail their payload check. Even an approved policy entry yields no outcome without its separately verified capability. A verifier can thus state that the pseudonymous keys jointly approved this exact tuple, but cannot infer who controls them without separate enrollment evidence.

The approval target is deliberately unsigned data until its detached signatures arrive. Timestamping it at that stage would establish only *target existence*: an attacker who later obtains a key could append a valid approval after the recorded time. Finalization therefore constructs an `acsd-approval-set/v1` object containing the target digest and the `SHA-256` digest of every exact author-approval COSE byte string, plus every separate predecessor-authorization signature used for an authority change. Only this closed object is the v0.3 timestamp subject. Replacing, adding, or removing a signature changes the approval-set digest. This is an evidence closure rule, not a new signature or timestamp primitive.

An exact predecessor digest proves structural continuity, not authority. A fresh-key attacker can copy a public WorkID, name the real predecessor, choose the consecutive version number, and validly sign the resulting child with the attacker’s keys. No hash or slot-conflict check answers whether those keys were entitled to create the parent-to-child edge. ACSD v3 therefore binds every release to a *lineage authority*: a sorted author-key set and a threshold for authorizing the next edge. The default threshold is unanimity. It is stored in the predecessor, so a child cannot lower it unilaterally.

$$\text{AuthorizedSuccessor}(P, C) = \text{ExactEdge}(P, C) \wedge \text{Authorization}(P, C),$$

where

$$\text{Authorization}(P, C) = \begin{cases} \text{Quorum}(A_P, \text{ApproveTarget}(C)), & A_P = A_C, \\ \text{Quorum}(A_P, \text{Transition}(P, C)) \wedge \text{AllApprove}(A_C, C), & A_P \neq A_C. \end{cases}$$

Here **ExactEdge** requires the same WorkID, exact parent release and PEC digests, and either a same-line increment by one or a new named line beginning at version one. **Transition** binds both authorities and the exact child release, governance, and PEC digests. The child approval target binds the transition digest. The construction is acyclic because the transition refers to complete child bodies while detached approvals refer to their completed digests.

For an unchanged authority object, the ordinary child approvals already constitute predecessor authorization and no duplicate signing step is needed. A key-set or threshold change uses double control: the predecessor threshold signs the exact transition and every new author signs the child target. This resembles the old-and-new authorization pattern used for TUF root-key updates, but ACSD applies it to a single scholarly lineage edge rather than software repository metadata [15]. A series key may advertise a verified child as a head; it cannot manufacture predecessor-author assent by itself.

Consider the concrete  $n+1$  attack. Mallory constructs a child with P’s WorkID, P’s exact release and PEC digests, the next version number, Mallory’s public key, and a valid Mallory signature over every child-controlled object. The object is internally signed but has neither an unchanged predecessor authority nor a transition signed by  $A_P$ . The implementation returns **VALID\_OBJECT\_BUT\_UNAUTHORIZED\_SUCCESSOR**. It does not accuse Mallory of plagiarism; it rejects only the claimed lineage edge.

This conclusion is relative to an accepted exact predecessor, not to the WorkID string alone. The CLI can pin a previously accepted parent with **-expected-parent-release-id**; otherwise it reports **UNPINNED\_EXACT\_PARENT**. Without an external parent anchor, an isolated package proves only that the authority named by its embedded parent authorized the edge; it cannot distinguish a separately fabricated genesis that copied a public WorkID.

Threshold choice exposes a real safety–liveness tradeoff. Unanimity prevents one coauthor from transferring the lineage alone but lets one lost or dissenting key freeze it. A lower threshold improves availability and is therefore allowed only when committed by the predecessor. There is no post-hoc identity recovery in this profile: without a precommitted recovery authority, lost threshold keys safely stop the cryptographic lineage. Nor can a later message erase a valid past authorization. Two valid distinct children in the same exclusive slot are reported as equivocation with no automatic winner; choosing by observation time requires an external transparency or governance policy.

### 3.4 Citation witnesses

The prototype recognizes only literal occurrences of a canonical token **urn:uuid:<work-id>** in the UTF-8 manuscript bytes. For every occurrence it records and signs

$$(\text{WorkID}, \text{byteOffset}, \text{byteLength}).$$

The published **reference\_work\_ids** list must equal the ordered de-duplicated projection of these witnesses. This prevents a series package from inventing a citation edge that has no corresponding token in its source release, and prevents a copied text from silently relabeling an old target as a new WorkID.

The rule intentionally does not parse PDFs, infer references from natural language, or determine whether a citation is intellectually appropriate. It is an exact byte-binding contract, not a semantic understanding claim.

For example, one UTF-8 source line contains the following token:

```
urn:uuid:12345678-1234-1234-1234-123456789abc
```

The witness records that WorkID, the byte offset of the first character of the token, and its byte length (45 bytes for this token). An editor that inserts a character before the token must regenerate the witness. This is intentionally mechanical: a source-level helper or build macro can insert the token, but v1 does not yet provide such editor integration, and hiding the token in a PDF would defeat this scanner.

A minimal release fragment is conceptually:

```
{ "slot": { "line": "main", "version": 1, "work_id": "urn:uuid:...",
  "content": { "sha256": "..."},
  "reference_work_ids": [ "urn:uuid:..." ],
  "citation_witnesses": [ { "work_id": "urn:uuid:...",
    "byte_offset": 35, "byte_length": 45 } ] }
```

The ellipses above are presentation placeholders only. The released demo fixture binds a real manuscript's bytes, digest, and work identifier (the governance and PEC objects are real, not placeholders), using synthetic author identifiers. Real series releases are produced by the v1 fixture generator.

## 4 Series Layer and Cyclic Citations

### 4.1 Two graphs, two purposes

A paper remains verifiable after all series packages are removed. The optional series layer records relationships among independently complete releases. A signed series package contains

$$B_k = \text{Sign}_S(\text{seriesId}, \text{epoch}, \text{sequence}, \text{previousBundleHash}, \\ \text{rotationHash}, \text{rank}, \text{releases}, \text{heads}, \text{semanticEdges}).$$

The semantic graph has vertices WorkIDs and may contain  $P \rightarrow Q$  and  $Q \rightarrow P$ . Its edges express declared citation or related-work relations. The commitment graph has series anchors and completed ReleaseIDs. It does not make a release depend on a future series package. Thus two authors can cite stable random WorkIDs before both final documents are content-addressed; a later package resolves each WorkID to a completed ReleaseID. The semantic graph may be cyclic while the commitment graph remains a DAG.

The package includes a signed structural rank strictly above every release version it resolves. Combining anchor-to-release edges with exact release-to-parent edges decreases this rank along a commitment path. This rank is a graph certificate, not a timestamp, publication date, or priority claim.

The package fields have narrow roles: *epoch* names the active series key generation; *rotationHash* commits its authorized key transition; *heads* maps each WorkID to its advertised line head; and *semanticEdges* records declared citation or related-work relations. The rank is only a finite measure certifying that the commitment graph has no cycle.

## 4.2 Revisions, branches, and conflicts

A revision names one exact predecessor. Multiple children of a parent are allowed, because a main manuscript and a computational appendix can be legitimate independent lines. A paper-level equivocation is instead detected when two valid, different releases occupy the same exclusive slot. A series-level equivocation is detected when two valid packages share the same series, epoch, sequence, and predecessor hash while committing to different contents.

An adversary can copy the content bytes, issue a fresh WorkID and keys, and obtain a valid new lineage. ACS D classifies this as content reuse under a different lineage, not as a cryptographic proof of plagiarism. That restraint is important: reuse may be authorized, collaborative, or independently derived.

As a concrete attack, suppose an attacker copies P-v1, changes its WorkID, and asks the series package to claim that it cites Q. The release verifier recomputes the content digest and byte witnesses; if the copied source does not contain the declared Q token, the package is rejected as an unsupported edge. If the attacker copies the token too, the package can still be valid under a new lineage. The evidence then says only “content reuse under a different lineage”; deciding plagiarism or authorization remains outside the verifier.

## 5 Scoped Post-release Statements

Some evidence is necessarily published after the immutable release: authors may open a committed dialogue window or reveal one name after review. Treating these as arbitrary metadata would reintroduce policy amplification. ACS D uses a small common rule for a statement  $E$  of type  $k$ , exact subject  $s$ , signer  $a$ , and requested claim  $c$ :

$$\begin{aligned} \text{Grant}(E, c) \Rightarrow & \text{SignatureValid}(E) \wedge \text{ScopeValid}(E, s) \\ & \wedge \text{Authorized}(k, s, a) \wedge c \in \text{Permitted}(k) \wedge \text{Compatible}(k, c). \end{aligned}$$

This shared skeleton is intentionally not a general policy language. Event, identity, and time evidence retain different schemas, authorization principals, and proof predicates.

For a dialogue event, the verifier derives the required signer set from the already approved PEC rather than trusting identifiers listed by the disclosure. It then checks the disclosure policy, exact PEC/event binding, contiguous Merkle opening, public-key identifiers, and every required COSE signature as one acceptance decision. A valid opening detached from its policy or a valid signature over a different opening grants nothing.

The slot-key identity assertion is an external per-slot sidecar binding the exact ReleaseID, WorkID, author slot, slot key, identity assertion, fixed purpose, and optional publication reference. Its result is deliberately named `SLOT_KEY_ASSENT_TO_IDENTITY_ASSERTION`: it says that the release slot’s pseudonymous key signed the mapping, not that a natural-person identity or venue acceptance was independently verified. One slot can be revealed without revealing another. A full-byline result requires a valid sidecar for every release slot; conflicting assertions for one slot have no protocol-selected winner.

The aggregate verifier labels any proper valid subset `PARTIAL_BYLINE_KEY_ASSENT`; it returns `FULL_BYLINE_KEY_ASSENT` only after quantifying over every slot in the accepted release. An optional fail-incomplete mode lets a publication workflow require full coverage without denying that partial disclosure can be intentional. A common signed publication-reference string can crosswalk the released bytes to a later record, but it remains an author-key assertion: it does not derive venue acceptance. Extra contribution fields are rejected rather than reinterpreted as contribution evidence.

Exact sidecar scope also does not create cross-paper unlinkability. Reusing the same public key in two unrelated WorkIDs exposes an equality edge even though a sidecar for one release cannot be replayed against the other. The read-only key-reuse audit reports this distinction; generating an independent key per unrelated lineage is the privacy-preserving default, not a proof against all traffic, account, or timing linkage.

## 5.1 Submission and unblinding lifecycle

A venue decision is deliberately not an ACSD state transition. Rejection can require no public action, leaving the previously released bytes unchanged. If review produces changed manuscript bytes, the submission becomes an exact predecessor-authorized successor. If the accepted text is byte-identical and only the byline is revealed, per-slot identity sidecars bind the existing release directly; manufacturing a redundant  $n+1$  version would conflate identity disclosure with scientific revision. In either case, the authors may sign a publication reference, but only independently authenticated venue evidence could establish receipt or acceptance.

## 6 Time Evidence

Time is intentionally external to author signing. An RFC 3161 request contains the SHA-256 imprint of the frozen approval set. A response supports the limited statement that the TSA signed a token for an imprint matching the locally recomputed complete target-and-signature set no later than its recorded `genTime`, only when the verifier supplies a TSA signer-certificate or fingerprint pin independently of the package. A certificate and its fingerprint copied into the same mutable package are not a trust anchor. The adapter checks the request/response nonce, SHA-256 imprint, `id-ct-TSTInfo` content type, single signer identifier, critical and exclusive timeStamping EKU, ESS certificate identifier, signed attributes, and CMS signature. It implements an exact-signer pin, not general PKIX path construction or revocation. Its local test TSA is a protocol test and can never produce `APPROVAL_SET_EXISTED_NOT_AFTER`.

There is no circularity: the timestamp receipt is a sidecar over the already closed approval set. A later revision can bind the receipt hash if a new signed object must refer to it. A timestamp does not prove creation at that moment, authorship, originality, or global first discovery. Multiple independently governed services can reduce single-service reliance, but their composition requires a separate policy.

Legacy releases timestamped only the approval target. They retain a distinct target-existence result and are never upgraded to the v0.3 approval-set result. This typed distinction closes a temporal-confusion attack while preserving backward verification.

## 7 Implementation and Evaluation

The v1 prototype uses Python for deterministic fixture generation and a zero-dependency Node.js verifier for strict CBOR/COSE and profile validation; the two paths are intentionally separate. The package is designed for offline replay: locked dependencies are included, and the verification driver regenerates fixtures, verifies a standalone release, independently verifies the series profile, builds a manifest, and verifies that manifest.

A command-line tool (`acsd.py`) turns the profile into an executable workflow. Core commands are `keygen`, `init`, `approve`, `finalize`, and `verify`. Lineage uses `authorize` and `revise`; a separate `compare-successors` command detects same-slot forks. Unblinding uses `disclose-identity` and

`verify-identity`; the aggregate command `verify-identity-set` distinguishes partial from full coverage. Finally, `audit-key-reuse` checks cross-work public-key equality. A one-command `release` path covers a locally held key set. Approval status is derived from verified COSE files, never trusted from mutable state metadata. COSE uses the registered EdDSA value `-8`, requires tag 18, and rejects non-minimal or ambiguous CBOR encodings. A zero-dependency Node path independently verifies Python-produced approvals. The canonical JSON layer is exercised by 64 fixed Python–Node vectors plus 1,000 seeded generated cases; all generated cases agree on canonical bytes and text-layer verdicts. Three known object-layer float typing differences do not change the text-layer rejection result and remain recorded in the canonical specification. The installable wheel and immutable paper/evidence directory are separate artifacts; the authoritative gate runs generated work in temporary directories and confirms source-tree byte identity afterward.

The negative corpus includes the inherited rank, unsupported-edge, citation-offset, canonical-JSON, and path attacks. New cross-layer regressions replace a public key under an old key id; coherently replace governance and PEC; recompute an unsigned target and attempt to reuse old approvals; alter COSE tags, algorithm identifiers, map keys, and lengths; replace a timestamp and rewrite the transport manifest; mismatch the nonce, signer, ECU, TSTInfo content type, or signature algorithm; and attempt to promote the local test TSA. All are rejected at their intended boundary. The single freeTSA result is an existence check that the adapter can consume one real service response, not a reliability study or a multi-provider conformance survey.

Additional regressions mutate approval-signature bytes after closure, attempt to relabel a target-only timestamp as approval-set time, detach a valid dialogue opening from its PEC or unanimous disclosure signatures, substitute a slot key, replay an identity mapping to another release, and promote a partial unblinding to a full byline. The deterministic demo now carries real Ed25519 release-approval and disclosure-approval signatures rather than self-reported signer identifiers. Its verifier reconstructs the release approval target, checks every required release approval, then verifies the separately scoped event disclosure against the accepted PEC digest.

The executable semantic-confusion challenge applies four verified evidence kinds to four claim kinds. Only the diagonal combinations are derivable; all twelve cross-kind substitutions are denied. Three deliberately unsupported social claims—natural-person identity, originality, and contribution truth—remain unconstructible in the closed derivation table. A separate timeline confirms that a target timestamp obtained before key compromise does not establish that approvals added after compromise existed at that earlier time.

Table 1: Typed evidence-to-claim confusion matrix. Checkmarks are the only derivable combinations; every off-diagonal cell is denied.

| Evidence                  | Target time | Set time | Registered | Slot assent |
|---------------------------|-------------|----------|------------|-------------|
| Approval-target timestamp | ✓           | –        | –          | –           |
| Approval-set timestamp    | –           | ✓        | –          | –           |
| SCITT inclusion           | –           | –        | ✓          | –           |
| Slot-identity assent      | –           | –        | –          | ✓           |

The lineage corpus begins with the smallest discriminating attack: a child with the exact WorkID and parent release/PEC digests,  $n+1$ , a fresh attacker key, and complete self-approval. It is rejected at the missing predecessor-quorum boundary. Positive cases cover unchanged authority, key rotation, one-of-two and two-of-two thresholds, and a named branch; negative cases cover threshold downgrade and transition-signature replay. Two valid same-slot siblings yield equivocation with no winner.

Table 2: Executable evaluation corpus. The first seven rows are inherited v1 fixture results; the remaining rows are reproducible from v3.

| Object or check                                         | Count / result                                            |
|---------------------------------------------------------|-----------------------------------------------------------|
| Standalone release versions                             | 8 / 8 valid without series packages                       |
| Author endorsements                                     | 18 / 18 bound to the intended release                     |
| Signed series objects                                   | 7                                                         |
| Closed normal, attack, and indeterminate scenarios      | 13 / 13 expected outcomes                                 |
| Independent profile checks                              | 30 / 30 passed                                            |
| Canonical JSON regression cases                         | 11 / 11 Python–Node agreement                             |
| Public-manifest entries                                 | 113 / 113 verified                                        |
| v3 differential canonical-JSON vectors                  | 64 / 64 accept/reject agreement                           |
| Python unit tests                                       | 163 passed; 3 environment-dependent tests skipped locally |
| Seeded generated canonical cases                        | 1000 / 1000 byte and verdict agreement                    |
| Typed semantic-confusion matrix                         | 16 / 16 expected decisions; 3 unsupported claims denied   |
| Independent approval implementations                    | Python producer / Node verifier                           |
| Authorized-lineage scenarios                            | 8 / 8 expected outcomes                                   |
| Submission lifecycle                                    | rejection, revision, and publication crosswalk reproduced |
| Multi-author unblinding                                 | 1/3, 2/3, 3/3 plus 3 adverse cases                        |
| Cross-work key isolation                                | replay rejected; reused key reported linkable             |
| Same-primitive lineage baseline                         | 3 policies; manual exact transition matches core decision |
| Real-TSA existence check                                | 1 freeTSA response verified                               |
| Strict scoped-certificate differential                  | 35 / 35 Python–Lean derivations                           |
| Authorized-lineage certificate differential             | 15 / 15 Python–Lean decisions                             |
| Production appraisal-transcript differential            | 19 / 19 Python–Lean decisions                             |
| PEC, lineage, scoped-appraisal, and transcript theorems | 84; no <code>sorry/admit</code>                           |

To avoid a weak comparison, a controlled adjacent baseline fixes the signature primitive to Ed25519 and evaluates three policies over the same parent and  $n+1$  candidates. Child-declared-key verification accepts both legitimate rotation and fresh-key capture; predecessor-key pinning rejects both. A manual predecessor signature over exact parent, child, and successor-key bytes correctly accepts the rotation and rejects replay onto the attacker child. ACSF reaches the same core decision. Its measured addition is therefore the maintained canonical object, threshold-governance, approval-closure, and typed claim profile, not a new signature or an assertion that hand-built transition statements are impossible.

Three further executable workflows test post-release use. A two-author submission lifecycle leaves a rejected release byte-identical, creates an authorized successor only when manuscript bytes change, and binds an unchanged release to a later publication reference without deriving acceptance.

A three-author unblinding run distinguishes 1/3 and 2/3 partial disclosure from 3/3 full coverage, and rejects duplicate slots, name mutation, and a smuggled contribution field. A cross-paper run rejects sidecar replay across releases while reporting that reuse of the same public key makes two WorkIDs publicly linkable.

The measurements remain prototype-scale rather than a throughput claim. They were recorded under Python 3.12.14 on 64-bit Windows 10.0.26200, a 13th Gen Intel Core i7-1360P with 16 logical processors, and 31.7 GiB physical memory. In that run, five repetitions of an in-memory workload that canonicalizes objects, constructs and verifies SHA-256 manifest entries, and scans a linear rank certificate had median times of 2.23, 26.45, 227.54, and 718.81 ms for 10, 100, 1,000, and 5,000 objects respectively; peak traced memory at 5,000 objects was 7.04 MiB. This excludes filesystem, COSE, and network cost, so it is a smoke test for obvious scaling anomalies, not a statistically powered performance evaluation; dispersion beyond minimum and median was not retained.

The v3 Lean development has 84 theorem statements over approval targets, approvals, digest-linked events, a closed outcome type, exact-subject scoped statements, lineage transitions, and claim-free appraisal transcripts. They prove, among other facts, that an approval for one target digest cannot be reused for a different target; every granted result has policy permission and compatible exact-subject evidence; all required approvals are present; and external time requires the approval-set subject, signer pin, nonce, RFC profile, and a non-local authority. The lineage theorems show that every accepted successor has predecessor-quorum evidence, a fresh authority cannot enter through the unchanged-authority path, and a transition bound to one child digest cannot authorize another child. The scoped layer proves that identity evidence cannot grant an event-opening claim, a missing slot prevents full-byline status, and a target-only timestamp cannot establish approval-set time. Evidence-set union cannot manufacture a capability: every derived claim still has compatible verified support in one component.

Unlike the earlier purely abstract boundary, Lean now strictly decodes the restricted canonical JSON for the maintained claim-free certificates and the generic `acsd-appraisal-transcript/v1` used by `acsd verify`. Its executable checker constructs exact-subject derivations, and refinement theorems map those derivations to the earlier digest-scoped abstraction for any chosen subject projection. Python and Lean agree on 35 scoped-certificate, 15 authorized-lineage, and 19 production-appraisal positive and adverse cases. The model deliberately accepts already checked signature, digest, and adapter facts as inputs: it does not refine filesystem loading, release-object parsing, DER/CBOR or COSE adapters, RFC 3161 processing, or the cryptographic primitives, and it does not prove TSA honesty or general PKIX correctness.

These 84 are PEC, lineage, scoped-appraisal, transcript, and cross-layer composition theorems, not a recount of the separately published v1 Lean core.<sup>1</sup> The v1 artifact records 53 theorem statements about release catalogs, teams, lineage, series graphs, citation witnesses, and lifecycle rules. That earlier model remains an inherited, distinct proof surface; it was neither deleted nor compressed into the new count. The v3 count concerns the executable composition model of approval-target reuse, capability gating, event links, typed external-time preconditions, selective unblinding, authorized succession, and strict appraisal-transcript checking.

## 8 Related Work

The profile builds on established foundations. PROV-DM supplies a general vocabulary for entities, activities, agents, derivations, and bundles [1]. Software Heritage identifiers provide intrinsic

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<sup>1</sup>The v1 proof surface is archived at <https://github.com/huiminZheng-collab/acsd-anonymous-series-v1/tree/main/formal-core>.

identifiers for software objects [2]; Trusty URIs provide hash-bearing immutable artifact identifiers [3]. COSE defines the signed binary envelope used for approvals [4]. SCITT specifies a broader architecture for signed statements and transparency services [5], while COSE Receipts standardizes receipt envelopes [6]; neither registration nor a valid receipt makes the registered statement true. RFC 3161 specifies the timestamp protocol [7]; RFC 9921 explicitly distinguishes COSE-then-timestamp from timestamp-then-COSE [8]. ACSD uses the former ordering as a semantic analogy: its separate approval-set object is timestamped after signature collection, without claiming wire-level interoperability with the RFC 9921 header profile. TUF uses predecessor and successor root thresholds to preserve trust through root-key updates [15]; in-toto binds supply-chain steps to authorized functionaries [16]. ACSD specializes related authorization ideas to an exact anonymous scholarly WorkID succession rule.

C2PA Content Credentials bind signed provenance assertions to media assets and explicitly separate tamper validation from judgments about whether an assertion is good or true [11]. The CAWG Identity Assertion adds creator identity assertions and validation results for that ecosystem [12]. ClaimChain provides privacy-preserving authenticated claim repositories with non-equivocation goals [13]. These systems demonstrate that signed provenance, scoped identity, selective disclosure, and conflict resistance are established design families. ACSD does not claim those concepts as new; it contributes a smaller scholarly-release profile whose closed claim algebra, approval completion, revision authority, mutual-citation separation, and executable counterexamples are evaluated together.

SD-JWT defines selective disclosure for claims in a signed JSON payload [9], while BBS-based verifiable-credential suites additionally target unlinkable derived proofs [10]. ACSD does not reproduce those credential features: its basic unblinding deliberately publishes a link from an already public release-slot key to one identity assertion. The smaller mechanism avoids an issuer/holder credential layer when public linkage is the goal, while the typed claim boundary leaves richer unlinkable credentials as a separable service level rather than an implicit security claim.

A detached signature already authenticates exact bytes under a chosen public key. Moreover, a minimal predecessor-signed transition over exact parent, child, and successor-key bytes can reproduce ACSD’s core key-rotation decision. Our controlled baseline confirms this positive result instead of comparing only against self-declared or permanently pinned keys. ACSD’s distinction is the integration of this edge with canonical scholarly objects, ordered and threshold author authority, signature-complete closure, scoped disclosures, and a closed output vocabulary.

ACSD does not claim to replace these systems or invent threshold delegation. Its narrow contribution is an executable scholarly profile that combines: self-contained anonymous paper releases; exact predecessor-authorized succession; a separate series layer for mutual citations; exact text-token witnesses; explicit conflict predicates; and counterexamples against common overclaims about signatures, local clocks, and copied content.

Table 3: Capability boundary (not a performance comparison).

| System                                | Primary capability                                                         | ACSD boundary                                                                                   |
|---------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| OpenTimestamps                        | Blockchain-anchored timestamp proofs                                       | Optional time sidecar; no author-role or citation semantics                                     |
| Software Heritage / SWHID             | Persistent intrinsic identifiers for archived software objects             | ACSD binds scholarly release roles, declarations and paper relations                            |
| Zenodo                                | Versioned archival record and DOI                                          | ACSD’s byte-level policy and pseudonymous team assent remain explicit                           |
| RFC 3161                              | TSA-signed message-imprint time evidence                                   | Exact not-after evidence only; never authorship or originality                                  |
| SCITT / COSE Receipts                 | Transparency registration architecture and signed receipt envelopes        | Optional registration evidence; statement truth and scholarly authority remain separate         |
| C2PA / CAWG                           | Asset provenance plus creator identity assertions                          | Broader media ecosystem; ACSD uses pseudonymous author slots and exact revision authority       |
| SD-JWT / BBS VC                       | Selective claim disclosure; BBS can add unlinkable derived proofs          | ACSD uses direct release-slot assent when public linking is intended                            |
| ClaimChain                            | Privacy-preserving authenticated claims with non-equivocation goals        | ACSD disclosures are public sidecars and its fork predicate does not provide a global view      |
| TUF / in-toto                         | Threshold key update and authorized software-supply-chain functionaries    | ACSD scopes old/new authority to exact scholarly child objects                                  |
| Detached signature + exact transition | Exact-byte key control and hand-built successor authorization              | Matches the core edge decision; application must define closure, thresholds, scopes, and claims |
| ACSD                                  | Release, governance, authorized revision, witness and claim-policy profile | Composes these boundaries while retaining backend trust assumptions                             |

## 9 Limitations and Future Work

Several extensions are intentionally deferred. First, relations across independently administered series require policy and an interoperable cross-series package model. Second, anti-equivocation evidence from transparency logs or gossip requires deployed service assumptions. Third, although Lean strictly decodes the restricted appraisal transcript, the filesystem, release-object, DER/CBOR, COSE, RFC 3161, canonicalization, hash, and signature adapters have not been refined into that model. Fourth, generated private keys are unencrypted PEM files; a production CLI needs encrypted or operating-system-backed storage. Fifth, exact TSA signer pinning does not replace PKIX path construction or revocation when a deployment requires those services. Long-lived evidence also needs an explicit renewal policy, such as renewed timestamps or an evidence-record scheme when algorithms or certificates age [17]. Sixth, v3 has no precommitted recovery authority: loss of enough predecessor keys freezes the lineage, while an already valid transfer cannot be retroactively

revoked. Seventh, the adjacent baseline is a controlled mechanism experiment, not an installed-client comparison with GnuPG, OpenTimestamps, Software Heritage, or Zenodo; no independent users or production deployment were measured. Finally, content-anonymous publication does not hide repository-account, network, timing, or public-key linkage, and richer citation/editor integration remains future work.

These are reasons to state the present result narrowly, not reasons to weaken the useful evidence it already preserves.

## 10 Conclusion

ACSD turns a vague “anonymous signed paper” idea into a limited but executable evidence profile. It binds exact released objects to pseudonymous-key approvals, distinguishes a signed object from a predecessor-authorized successor, supports ordered roles and branches, permits mutual scholarly citation without cyclic hash dependencies, and detects same-slot authorized forks without inventing a winner. Post-release sidecars support voluntary partial or complete byline disclosure without converting an author-key publication crosswalk into venue acceptance. The positive manual-transition baseline also shows where the contribution stops: predecessor authorization can be assembled from ordinary signatures, while ACSD supplies one closed, threshold-aware, replay-tested profile. It leaves human identity, originality, recovery without prior authority, and independent time as claims requiring additional evidence. This division of claims is the central design principle: preserve what cryptography can verify, and make the remaining social questions explicit.

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